



## Neural Networks and Training



**Video:** Introduction to Neural Networks  
15 min



**Video:** Model Training in Edge Impulse  
7 min



**Reading:** Neural Networks and Training  
5 min



**Practice Quiz:** Neural Networks and Training  
5 questions

## Model Evaluation



**Video:** How to Evaluate a Model  
10 min



**Video:** Underfitting and Overfitting  
6 min



**Reading:** Evaluation, Underfitting, and Overfitting  
5 min



**Practice Quiz:** Evaluation, Underfitting, and Overfitting  
5 questions

## Deploying a Model



**Video:** How to Use a Model for Inference  
6 min



**Video:** Testing Inference with a Smartphone  
3 min



**Video:** How to Deploy a Trained Model to Arduino  
9 min



**Reading:** Using a Model for Inference  
5 min



**Practice Quiz:** Deploy Model to Embedded System  
5 questions



# Neural Networks and Training

While optional, we encourage you to take a look at the following articles and videos to learn more about the topics covered in this lesson:

Another introduction to neural networks, if you'd like another perspective: <https://www.youtube.com/watch?v=aircAruvnKk>

Here is a full (1+ hour) lecture on how neural networks work with forward- and backpropagation. It is a real deep dive and includes the calculus necessary for backpropagation: <https://www.youtube.com/watch?v=d14TUNcbn1k>

[Machine Learning for Beginners: An Introduction to Neural Networks](#)

[Using neural nets to recognize handwritten digits](#)

[A Gentle Introduction to the Rectified Linear Unit \(ReLU\)](#)

[A Simple Explanation of the Softmax Function](#)

✓ Complete

Go to next item

